

ORIGINAL RESEARCH

Fluid Management of Acute Heart Failure With the Reprieve System

The Randomized Controlled FASTR Trial

James E. Udelson, MD,^a Ali Javaheri, MD, PhD,^{b,c} Marat Fudim, MD, MHS,^d Kevin Damman, MD, PhD,^e Jan Biegus, MD, PhD,^{f,g} Aasim Afzal, MD, MBA,^h Alpesh N. Amin, MD, MBA,ⁱ Daniel Bensimhon, MD,^j Jason J. Bischof, MD,^k Eugene S. Chung, MD,^l Robert L. Gottlieb, MD, PhD,^{m,n,o,p,q} Simon A. Mahler, MD, MS,^r Jonathan D. Moreno, MD, PhD,^b Piotr Ponikowski, MD,^s Veena Rao, PhD,^t Zachary Cox, PHARM.D,^u Juan Ivey-Miranda, MD,^{v,w} Sean Collins, MD,^{x,y} Jeffrey Testani, MD, MS^{t,*}

ABSTRACT

BACKGROUND The Reprieve System is designed to overcome barriers limiting safe and rapid decongestion with individualized automated diuretic titration, real-time diuretic response monitoring, and individualized sodium chloride replacement to prevent cardio-renal dysfunction.

OBJECTIVES This study aims to establish proof-of-concept that the Reprieve System can facilitate rapid and safe decongestion.

METHODS FASTR (Fluid Management of Acute Decompensated Heart Failure Subjects Treated With Reprieve Decongestion Management System [DMS]) was a randomized pilot trial comparing the Reprieve System vs a control strategy of optimal diuretic therapy (ODT) in hospitalized patients with acute heart failure. The primary efficacy endpoint was 24-hour natriuresis, and the primary safety endpoint was a composite of dialysis or doubling of creatinine levels, severe electrolyte abnormalities, hypotension, or hypertensive emergency.

RESULTS A total of 100 patients were enrolled, with 96 receiving randomized treatment (Reprieve, n = 52; ODT, n = 44). At baseline, the median estimated glomerular filtration rate was 49 mL/min/1.73 m² (Q1-Q3: 36-78 mL/min/1.73 m²) with estimated excess fluid volume of 20 lbs (Q1-Q3: 15-35 lbs). Twenty-four-hour natriuresis was significantly greater with the Reprieve System (1,082 ± 487 mmol) vs ODT (423 ± 290 mmol; *P* < 0.001). The safety endpoint occurred in 31% of the Reprieve group vs 39% of the ODT group (*P* = 0.42). Intravenous diuretic therapy duration was shorter with Reprieve [46 hours (Q1-Q3: 29-80 hours)] vs ODT (88 hours [Q1-Q3: 44-143 hours]; *P* = 0.014). The rate of weight loss (*P* = 0.002), net fluid loss (*P* = 0.03), and net natriuresis (*P* < 0.001) were significantly faster with Reprieve. Change in serum creatinine levels did not differ between the Reprieve (0.19 ± 0.24 mg/dL) and ODT (0.31 ± 0.39 mg/dL; *P* = 0.07) groups.

CONCLUSIONS In this pilot trial, the Reprieve System safely produced significantly faster decongestion compared with ODT. Confirmation of these findings in the ongoing pivotal trial is required. (Fluid Management of Acute Decompensated Heart Failure Subjects Treated With Reprieve Decongestion Management System [DMS] [FASTR]; [NCT05174312](https://doi.org/10.1016/j.jchf.2026.103062)) (JACC Heart Fail. 2026;■:103062) © 2026 Published by Elsevier on behalf of the American College of Cardiology Foundation.

ABBREVIATIONS AND ACRONYMS

AHF = acute heart failure

CA-UTI = catheter-associated urinary tract infection

GDMT = guideline-directed medical therapy

IV = intravenous

NaCl = sodium chloride

ODT = optimal diuretic therapy

The primary therapeutic goal in the majority of acute heart failure (AHF) hospitalizations is decongestion.^{1,2} Despite this goal, the intensity and approach to diuretic therapy are highly variable.^{3,4} Poor fidelity of information for monitoring diuretic response, slow and inadequate diuretic dosing and titration, diuretic resistance, and worsening kidney function are common.⁵ Ultimately, these factors can synergize, leading to slow and incomplete decongestion.

To address these limitations, the Reprive System was designed around a 3-part strategy: 1) quantitation of diuretic response with high fidelity and in real-time; 2) automated dosing/titration of intravenous (IV) furosemide based on diuretic response; and 3) personalized sodium chloride (NaCl) replacement to prevent diuretic resistance and worsening kidney function while ensuring a negative sodium balance.⁶ Although the utility of accurate diuretic monitoring and timely titration may be intuitive, historically, NaCl administration during AHF treatment has been considered counterproductive. Multiple independent lines of current evidence, however, indicate that NaCl supplementation may have beneficial effects on sodium avidity and diuretic resistance, neurohormonal activation, and possible protection of kidney function.^{7–11}

Although the primary goal of diuretic therapy is to remove sodium, chloride is a critical regulatory factor for neurohormonal activation and renal sodium transport.¹² Notably, the primary molecular sensor for NaCl is the WNK-kinase system, which senses chloride levels and modulates renal sodium handling in response.^{13,14} Thus, by combining real-time information on diuretic response, rapid titration of loop diuretics, and personalized NaCl replacement, we tested the hypothesis that very rapid decongestion may be possible without the renal and metabolic disadvantages often seen with other aggressive diuretic strategies.^{15–18}

METHODS

PATIENT POPULATION. FASTR (Fluid Management of Acute Decompensated Heart Failure Subjects Treated With Reprive Decongestion Management System [DMS]; [NCT05174312](#)) was a randomized, prospective, open-label pilot trial. Its primary goal was optimizing the Reprive System algorithm and providing proof-of-concept that the Reprive System can facilitate rapid and safe decongestion compared with a control strategy of optimal diuretic therapy (ODT).

Patients ≥ 18 years of age hospitalized with a diagnosis of acute-on-chronic heart failure as defined by the presence of at least 1 symptom (dyspnea,

From the ^aDivision of Cardiology and the CardioVascular Center, Tufts Medical Center, Boston, Massachusetts, USA; ^bDivision of Cardiology, Department of Medicine, Washington University School of Medicine, St. Louis, Missouri, USA; ^cJohn Cochran VA Hospital, St. Louis, Missouri, USA; ^dDuke University Medical Center, Division of Cardiology, Duke Clinical Research Institute, Durham, North Carolina, USA; ^eUniversity of Groningen, University Medical Center Groningen, Groningen, the Netherlands; ^fDepartment of Cardiology, Clinical Department of Intensive Cardiac Care, Wroclaw Medical University, Faculty of Medicine, Institute of Heart Diseases, Wroclaw, Poland; ^gJan Mikulicz Radecki University Hospital in Wroclaw, Poland; ^hTexas A&M University College of Medicine and Baylor Scott and White Health, Plano, Texas, USA; ⁱDepartment of Medicine, Division of Hospital Medicine and Palliative Medicine, University of California, Irvine, California, USA; ^jCone Health Heart and Vascular Center, Greensboro, North Carolina, USA; ^kDepartment of Emergency Medicine, The Ohio State University College of Medicine, Columbus, Ohio, USA; ^lThe Lindner Research Center at The Christ Hospital, Cincinnati, Ohio, USA; ^mDepartment of Internal Medicine, Baylor University Medical Center, Dallas, Texas, USA; ⁿBaylor Scott and White, The Heart Hospital, Dallas, Texas, USA; ^oBaylor Scott and White, The Heart Hospital, Plano, Texas, USA; ^pDepartment of Internal Medicine, Burnett School of Medicine at TCU, Fort Worth, Texas, USA; ^qDepartment of Internal Medicine, Texas A&M Health Science Center, Dallas, Texas, USA; ^rDepartments of Emergency Medicine, Epidemiology and Prevention, and Implementation Science, Wake Forest University School of Medicine, Winston Salem, North Carolina, USA; ^sInstitute of Heart Diseases, Wroclaw Medical University, Wroclaw, Poland; ^tDepartment of Internal Medicine, Section of Cardiovascular Medicine, Yale University School of Medicine, New Haven, Connecticut, USA; ^uDepartment of Pharmacy Practice, Lipscomb University College of Pharmacy, Nashville, Tennessee, USA; ^vHospital de Cardiologia, Instituto Mexicano del Seguro Social, Mexico City, Mexico; ^wDepartment of Internal Medicine, Section of Cardiovascular Medicine, Yale University School of Medicine, New Haven, Connecticut, USA; ^xDepartment of Emergency Medicine and Veterans Affairs Tennessee Valley Healthcare System, Geriatric Research, Education and Clinical Center (GRECC), Nashville, Tennessee, USA; and the ^yDepartment of Medicine, Vanderbilt University Medical Center, Nashville, Tennessee, USA. *Dr Testani's contribution to this publication was made in their individual capacity and not as part of their Yale University duties or responsibilities.

The authors attest they are in compliance with human studies committees and animal welfare regulations of the authors' institutions and Food and Drug Administration guidelines, including patient consent where appropriate. For more information, visit the [Author Center](#).

Manuscript received February 9, 2026; revised manuscript received March 17, 2026, accepted March 18, 2026.

orthopnea, or edema/swelling) and 1 sign (peripheral edema, ascites, or jugular venous distension) and ≥ 10 lbs (4.5 kg) of estimated volume overload (as assessed by the treating clinician) who were chronically prescribed loop diuretic agents (minimum 30 days) were eligible. Exclusion criteria included inability to place a urinary catheter or peripheral IV catheter, hemodynamic instability (including systolic blood pressure < 90 mm Hg), an estimated glomerular filtration rate < 20 mL/min/1.73 m², or current use of renal replacement therapy, or severe electrolyte abnormalities (eg, serum potassium < 3.0 mEq/L, magnesium < 1.3 mEq/L, or sodium < 125 mEq/L). *Note: These are based on baseline/screening labs. Subjects whose electrolyte levels are repleted cannot be reassessed for inclusion in the trial.* A full list of the inclusion and exclusion criteria are provided in [Supplemental Table 1](#).

Participants were enrolled in the United States with oversight by local and central institutional review boards. All participants randomized to treatment provided written informed consent.

INTERVENTION AND CONTROL STRATEGIES. The Reprieve System is a bedside fluid management system consisting of a physical device and a novel therapeutic algorithm to provide personalized and automated infusion of IV furosemide and normal saline based on the patient's diuretic response. An image of the Reprieve System is provided in the [Supplemental Methods](#).

Diuresis was measured in real time and with high fidelity via a proprietary, primed urinary catheter-based urine collection system enabled by an internal scale. This urine collection system provides significant advantages over traditional urinary catheter collection systems as it ensures a continuous column of fluid between the bladder and urine measurement system, eliminating air locks and providing a 1:1 temporal relationship between urine production and urine volume measurement. The Reprieve system is designed to improve the fidelity and timeliness of diuresis monitoring, rapidly identify a therapeutic diuretic dose, and continually monitor urinary response to prompt adjustment to diuretic therapy as diuresis changes over time. It also incorporates a saline (0.9% NaCl) repletion algorithm with a goal of minimizing kidney dysfunction and preventing diuretic resistance while ensuring a net negative sodium balance.

The Reprieve system is designed to operate across the spectrum of care settings in which patients with AHF are usually treated (eg, emergency department, general hospital ward, intensive care unit) without

increasing nursing or other care requirements beyond usual care. The Reprieve System algorithm operates in 3 phases: 1) a dose-finding phase; 2) a continuous infusion phase; and 3) a therapy transition phase. These phases are described in the [Supplemental Methods](#).⁶

The major modification of the Reprieve algorithm during the trial was the transition from a urine volume to a natriuresis (sodium output)-based algorithm, which was implemented after 54 patients were enrolled. Early pilot work outside of the United States observed high urine sodium concentrations with little interpatient variability, suggesting that the incorporation of urine sodium into the algorithm might be unnecessary. However, at the planned interim analysis of FASTR, urine sodium concentrations were lower and had substantially greater interpatient variability than observed in the pilot work. This observation, in conjunction with the emerging evidence supporting natriuresis-guided therapy, drove the decision to change to a natriuresis-based algorithm.^{18,19} The resulting primary procedural difference is that when urine is emptied from the Reprieve System (each ~ 2 L of urine volume), a urine sample of this urine collection is sent to the local hospital laboratory for urine sodium concentration measurement. The resulting urine sodium concentration is manually entered into the Reprieve System. The natriuresis-based algorithm is essentially the same as the prior urine volume-based algorithm. The exception is that rather than assuming a constant high urine sodium concentration throughout therapy, the natriuresis-guided algorithm uses the patient's measured urine sodium concentration in combination with the measured urine volume to guide both IV furosemide and replacement NaCl infusions.

ODT CONTROL. Because there is no consensus for which aggressive diuretic strategy in AHF is superior, the control ODT strategy allowed clinicians to use best practices of diuretic dosing based on the recent DOSE (Diuretic Strategies in Patients with Acute Decompensated Heart Failure), ADVOR (Acetazolamide in Decompensated Heart Failure with Volume Overload), and CLOROTIC (Combination of Loop Diuretics with Hydrochlorothiazide in Acute Heart Failure) randomized clinical trials.¹⁵⁻¹⁷ The guidance given to clinicians for ODT is shown in [Supplemental Table 2](#). Given that the inclusion criteria for the current trial were different from that of the aforementioned trials (eg, $\sim 1/3$ of patients would have been disqualified from the DOSE trial due to too high baseline loop diuretic doses), the

investigator was allowed latitude to establish an aggressive “optimal” strategy based on one or a combination of the strategies used in these trials. Overall, this strategy was effective in producing an aggressive control therapy compared with contemporary trials ([Supplemental Figure 1](#)).

TRIAL ENDPOINTS. The primary objective of the study was to determine whether the Reprieve System could more efficiently relieve congestion in patients with AHF with similar or improved safety compared with ODT. The randomized diuretic therapies in the ODT and Reprieve treatment arms were continued until either the patient’s optimal volume status was achieved or completion of 72 hours of therapy after treatment initiation, whichever occurred first. After 72 hours, diuretic therapies were at the discretion of the provider in both arms.

The primary efficacy endpoint was natriuresis (total urine sodium output) 24 hours after treatment initiation. The primary safety endpoint, determined in cooperation with the U.S. Food and Drug Administration, was the composite of: 1) clinically significant acute kidney injury, defined as KDIGO (Kidney Disease: Improving Global Outcomes) stage 2 or greater acute kidney injury [doubling or more of baseline serum creatinine or use of renal replacement therapy]; 2) severe electrolyte abnormality (serum potassium <3.0 mEq/L, magnesium <1.3 mEq/L, or sodium <135 mEq/L); 3) symptomatic hypotension (systolic blood pressure <80 mm Hg with symptoms that require fluid resuscitation or vasopressor therapy); or 4) hypertensive emergency (blood pressure $>180/120$ mm Hg with evidence of end-organ damage).

Secondary efficacy endpoints were net fluid loss at the end of therapy and time on IV loop diuretics, defined as the time from initiation of randomized therapy to the last dose of the IV loop diuretic administered for AHF. The rate of decongestion was assessed as the diuretic response (ie, natriuresis, net fluid loss, weight change) achieved per 24 hours of treatment.

An independent Clinical Events Committee evaluated adverse events and adjudicated event severity and relatedness to the treatment or device. The Clinical Events Committee also adjudicated adverse events by using the Centers for Disease Control National Healthcare Safety Network criteria for catheter-associated urinary tract infections (CA-UTIs).²⁰

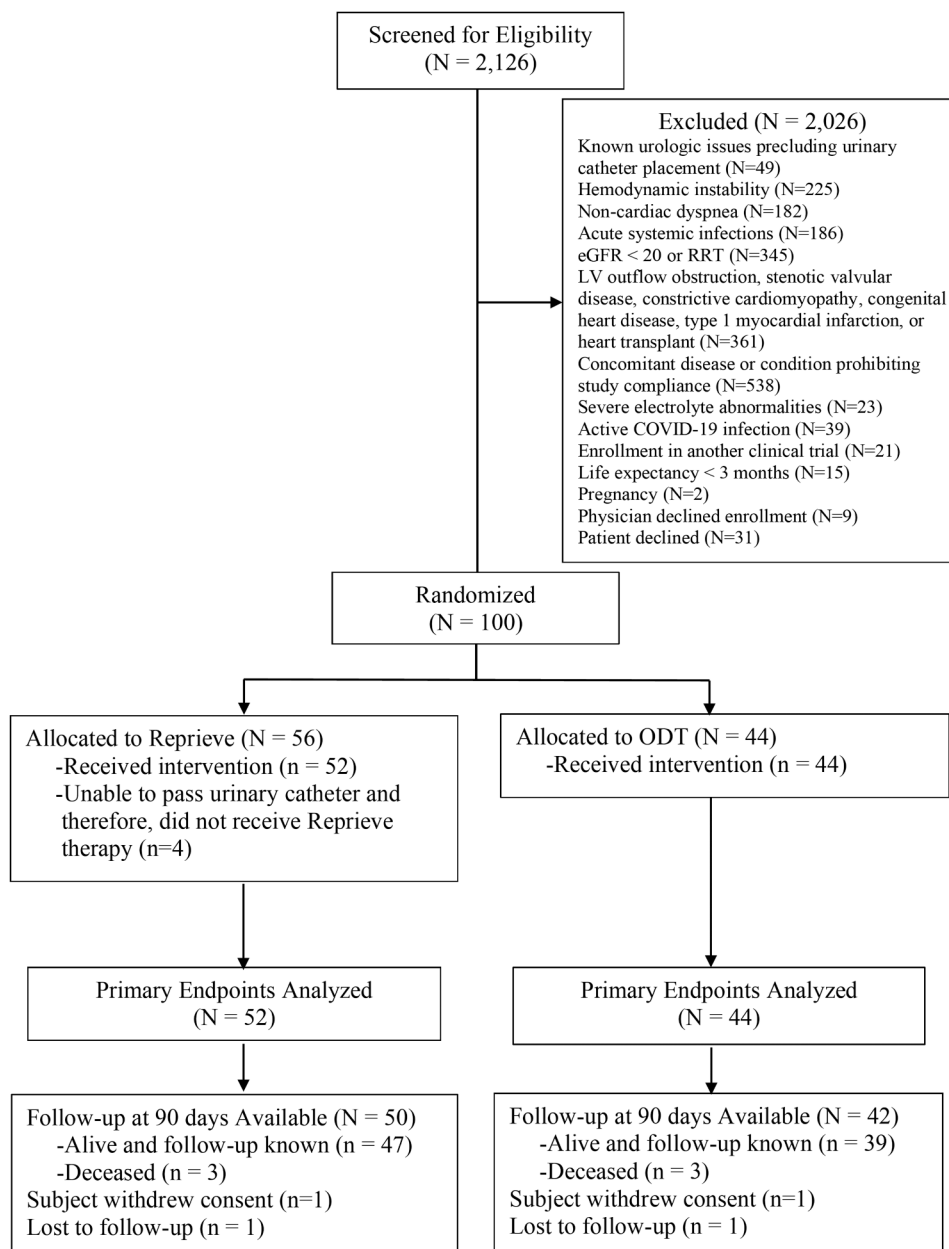
In addition to assessing ototoxicity as a clinically recognized adverse event during the study period, subclinical changes in hearing were assessed prospectively by formal audiometry performed at baseline, hospital discharge, and 7 days after discharge ([Supplemental Methods](#)).

CALCULATIONS. Loop diuretic doses were converted to furosemide equivalents with 1 mg oral/IV bumetanide = 20 mg oral torsemide = 40 mg IV furosemide = 80 mg oral furosemide.^{21,22} The estimated glomerular filtration rate was calculated by using the 2021 CKD-EPI (Chronic Kidney Disease Epidemiology Collaboration) equation.²³ Diuretic efficiency was calculated as cumulative natriuresis normalized for the log base 2 of the cumulative diuretic dose as previously validated.²⁴

STATISTICAL ANALYSIS. Because this was a pilot trial with a primary goal of optimizing the Reprieve System algorithm and providing pilot data to plan a future pivotal trial, there was no formal sample size assessment or priori statistical analysis plan for hypothesis testing. As such, all hypothesis testing is post hoc and was not adjusted for multiplicity. Thus, findings should be treated as exploratory, and nominal values of $P < 0.05$ should not be interpreted as confirmatory evidence. Normality was assessed graphically by means of histograms and Q-Q plots. Normally distributed variables are presented as mean $\pm$ SD and were compared by using the Student’s *t*-test. Skewed data are presented as median (Q1-Q3) and were compared with the Mann-Whitney *U* test. Categorical data were compared by using the chi-square test.

Variability between the urine output-based and natriuresis-based algorithms were compared by using the *F* test for the homogeneity of variances. Linear regression was used to evaluate if there were effect modifications between the natriuresis-based and urine output-based algorithms. Treatment arms, algorithms, and their interaction term were included as main effects. Time to loop diuretic titration or time to use of adjuvant thiazides is presented according to the Kaplan-Meier method.

For the analysis of time to further loop diuretic titration and time to adjuvant thiazide, patients who did not experience the outcome were censored at 72 hours. Patients who received upfront thiazides were analyzed as receiving the drug at time 0 and were included in the analysis. For the analysis of time to 90-day death or heart failure hospitalization, patients who did not experience the outcome were censored at the last follow-up visit; this model included first and subsequent events, and thus robust SEs were specified. Cox proportional hazards regression was used to analyze the 3 models. No covariables were included in the models of time to diuretic titration or adjuvant thiazides, but the model of 90-day death or heart failure hospitalization analysis was adjusted by the protocol version. In all models, the proportional hazards assumption was

FIGURE 1 Consort Flow Diagram

eGFR = estimated glomerular filtration rate; LV = left ventricular; ODT = optimal diuretic therapy; RRT = renal replacement therapy.

checked on the basis of Schoenfeld residuals and visually by plotting the $-\ln[-\ln(\text{survival})]$ curves for each category of the arm vs the $\ln(\text{time})$.

Statistical significance was defined as 2-tailed $P < 0.05$ without adjustment for multiple testing. Statistical analysis was performed with IBM SPSS Statistics version 26 (IBM SPSS Statistics, IBM Corporation) and Stata SE version 18.0 (StataCorp).

RESULTS

BASELINE CHARACTERISTICS. In total, 2,216 patients were screened between July 2022 and October 2024 at 16 sites in the United States (Figure 1). Subsequently, 100 patients were randomized; 96 patients received randomized study decongestive treatment (52 Reprive, 44 ODT) and comprised the

TABLE 1 Baseline Characteristics

	ODT (n = 44)	Relieve System (n = 52)	P Value
Age, y	63 ± 14	64 ± 13	0.87
Male	24 (55)	29 (56)	0.90
Race			
White	26 (59)	31 (60)	0.35
Black	18 (41)	18 (35)	
Other	0 (0)	3 (6)	
BMI, kg/m ²	42.9 ± 14.9	39.9 ± 8.2	0.44
LVEF, % ^a	37 ± 20	39 ± 18	0.60
LVEF >50%	13 (30)	14 (27)	0.77
Ischemic cardiomyopathy	39 (89)	47 (90)	0.78
Estimated excess fluid, lbs	24 (15-40)	20 (15-35)	0.76
Comorbid conditions			
Hypertension	39 (89)	45 (87)	0.76
Diabetes mellitus	28 (64)	33 (64)	0.99
History atrial fibrillation/flutter	21 (48)	28 (54)	0.55
Vital signs			
Systolic blood pressure, mm Hg	122 ± 21	117 ± 17	0.17
Diastolic blood pressure, mm Hg	69 ± 14	68 ± 12	0.61
Heart rate, beats/min	78 ± 17	79 ± 15	0.33
Medications			
ACEI or ARB or ARNI	23 (52)	20 (38)	0.18
Beta-blocker	32 (73)	28 (54)	0.06
Mineralocorticoid receptor antagonist	21 (48)	26 (50)	0.82
SGLT2 inhibitor	21 (48)	23 (44)	0.73
Home daily oral loop diuretic in IV FE, mg	60 (35-160)	80 (40-140)	0.56
Laboratory values at therapy initiation			
Serum sodium, mEq/L	137.5 ± 3.5	137.6 ± 3.0	0.89
Serum chloride, mEq/L	98.8 ± 5.7	100.2 ± 4.6	0.21
Serum bicarbonate, mEq/L	28.7 ± 4.0	27.5 ± 4.3	0.17
Serum creatinine, mg/dL	1.44 ± 0.56	1.48 ± 0.56	0.74
Blood urea nitrogen, mg/dL	28 ± 15	29 ± 17	0.75
eGFR, mL/min/1.73 m ²	57 ± 25	55 ± 23	0.60
NT-proBNP, pg/mL ^a	2,256 (682-5,315)	3,051 (918-5,989)	0.43

Values are mean ± SD, n (%), or median (Q1-Q3), unless otherwise indicated. ^aData only available in 78 patients.

ACEI = angiotensin-converting enzyme inhibitor; ARB = angiotensin II receptor blocker; ARNI = angiotensin receptor-neprilysin inhibitor; BMI = body mass index; eGFR = estimated glomerular filtration rate; FE = furosemide equivalents; LVEF = left ventricular ejection fraction; NT-proBNP = N-terminal pro-B-type natriuretic peptide; ODT = optimal diuretic therapy; SGLT2 = sodium-glucose cotransporter 2.

TABLE 2 Adjunctive Diuretics

	ODT (n = 44)	Relieve System (n = 52)	P Value
Thiazide during the first 24 h	10 (22.7)	29 (55.8)	0.001
Thiazide between 24 h and 72 h	4 (9.1)	3 (5.8)	0.53
Thiazide overall	14 (31.8)	32 (61.5)	0.004
Acetazolamide during the first 24 h	2 (4.6)	2 (3.9)	0.86
Acetazolamide between 24 h and 72 h	5 (11.4)	2 (3.9)	0.16
Acetazolamide overall	7 (15.9)	4 (7.7)	0.21

Values are n (%), unless otherwise indicated.

Abbreviation as in Table 1.

analytical cohort. Major reasons for exclusion before randomization included infection, advanced kidney dysfunction, hemodynamic instability, and concomitant diseases.

Overall, baseline characteristics were balanced between treatment arms (Table 1). The mean age was 63 ± 14 years, 55% were male, and the median estimated glomerular filtration rate was 49 mL/min/1.73 m² (Q1-Q3: 36-78 mL/min/1.73 m²). Patients were well treated at baseline with contemporary guideline-directed medical therapy (GDMT). Preadmission oral loop diuretic doses were high, with an average dose of 188 ± 158 mg/day of oral furosemide equivalents (median 160 mg/day [Q1-Q3: 80-320 mg/day]). Patients had a median of 20 lbs (Q1-Q3: 15-35 lbs) of estimated excess fluid volume at randomization.

DECONGESTIVE THERAPIES. Overall, ODT diuretic dosing was intense, with 52% of patients receiving ≥2.5 times the home dose of IV loop diuretic and 38.6% of patients receiving either adjuvant thiazide or acetazolamide during the study period (Table 2). In the first hour of therapy, patients in the ODT group received a median dose of 22 mg (Q1-Q3: 10-80 mg) of IV furosemide compared with 145 mg (Q1-Q3: 72-201 mg) in the Relieve System group (Figure 2). The low up-front furosemide exposure in ODT was driven largely by the use of continuous infusion (59% of ODT initial therapy; median starting dose 20 mg/h) without a starting bolus dose.

During therapy, 23 patients had a loop diuretic up-titration with the Relieve System (82.1% of those who were not on the maximal infusion rate after the dose-finding phase and thus eligible for up-titration) compared with 14.6% of ODT patients ($P < 0.001$) (Figure 2). The median time to titration was also longer with ODT (19.5 hours [Q1-Q3: 15-46 hours]) compared with Relieve (11.1 hours [Q1-Q3: 6.8-15.8 hours]; $P = 0.046$). Similarly, the proportion of patients receiving adjuvant thiazide was substantially higher in the Relieve group (61.5%) compared with the ODT group (31.8%; $P = 0.004$). Overall, the cumulative IV loop diuretic dose over the 72-hour study period was 1,342 mg (Q1-Q3: 1,012-2,182 mg) with Relieve compared with 480 mg (Q1-Q3: 240-708 mg) furosemide equivalents with ODT ($P < 0.001$). The Relieve System administered an average of 3.3 ± 1.7 L/day IV normal saline replacement during the treatment period.

PRIMARY ENDPOINTS. The primary efficacy endpoint of natriuresis at 24 hours was 1,082 ± 487 mmol of urine sodium with Relieve and 423 ± 290 mmol of

sodium with ODT ($P < 0.001$) (Central Illustration). The primary composite safety endpoint occurred in 31% of patients in the Reprieve arm vs 39% of patients in the ODT arm ($P = 0.42$) (Table 3). With respect to the components of the primary safety composite, no Reprieve patients and 2 ODT patients experienced a doubling of serum creatinine levels during the treatment period. Renal replacement therapies were not required in either arm.

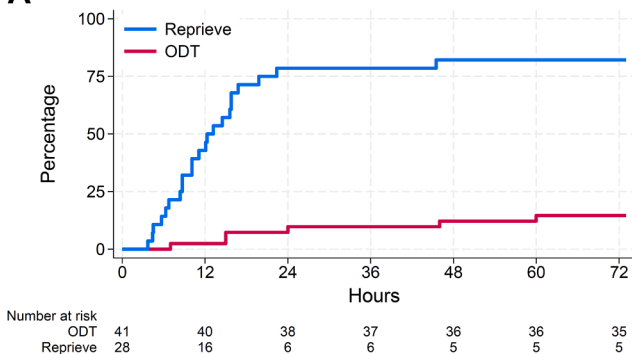
One or more of the specified electrolyte abnormalities occurred in 31% with Reprieve and 36% with ODT. The majority of electrolyte abnormality safety events were serum sodium concentrations falling below 135 mEq/L, with few severe hypokalemia or hyponatremia events in either arm (Table 2). When the primary composite safety endpoint was analyzed at 24 hours, it occurred in 11.54% of patients in the Reprieve arm vs 11.36% of patients in the ODT arm ($P = 0.92$). The favorable balance of the primary efficacy and safety endpoints were similar at 24 hours and end of therapy (Supplemental Figure 2).

SECONDARY AND EXPLORATORY OUTCOMES. Patients in the ODT arm had a significantly longer total time on IV diuretic agents (comprising both randomized and open-label IV diuretic therapy) of 88 hours (Q1-Q3: 44-143 hours) compared with 46 hours (Q1-Q3: 29-80 hours) in patients treated with Reprieve ($P = 0.014$). During the 72-hour study period, 71% of patients in the Reprieve arm completed IV diuretic therapy compared with 41% in the ODT arm ($P = 0.003$). Despite Reprieve reducing the IV diuretic therapy time by nearly 50%, an equivalent net fluid loss (8.1 ± 5.7 L vs 8.1 ± 7.1 L; $P = 0.98$), net sodium excretion (841 mmol [Q1-Q3: 545-1,485 mmol] vs 890 [Q1-Q3: 410-1,518 mmol]; $P = 0.81$), and weight loss (6.6 ± 4.1 kg vs 5.6 ± 4.6 kg; $P = 0.27$) were achieved during the 72-hour study period with Reprieve vs ODT, respectively. As such, the rate of decongestion across multiple metrics was substantially improved with Reprieve vs ODT (Figure 3).

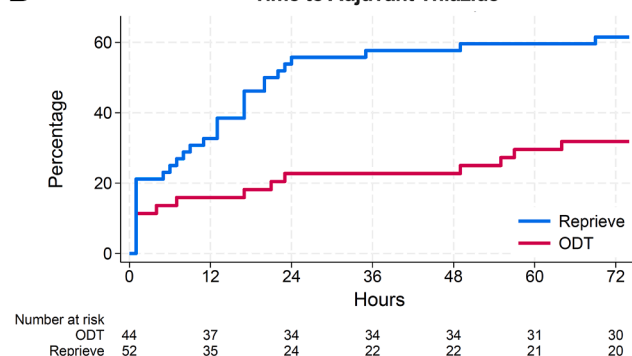
Although the primary efficacy endpoint of cumulative natriuresis remained positive at 72 hours, attenuation of difference on the secondary and exploratory efficacy endpoints was observed at that time point (Supplemental Figure 2). Natriuresis and diuresis did not differ between patients with and without a urinary catheter in the ODT arm ($P > 0.27$). Overall, Reprieve therapy had a favorable effect on loop diuretic resistance as diuretic efficiency

FIGURE 2 Diuretic Therapies and Titration

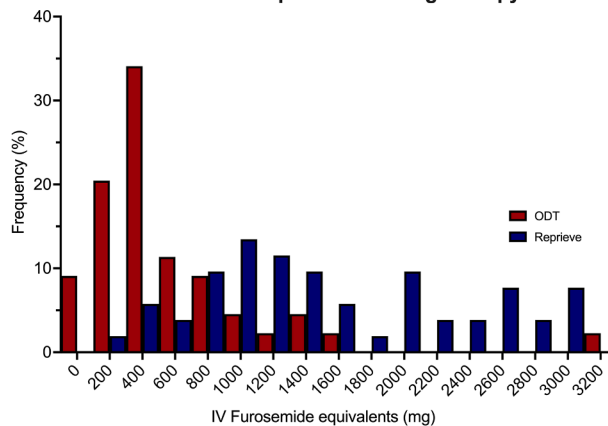
A Time to Further IV Loop Diuretic Titration after Dose Finding Phase



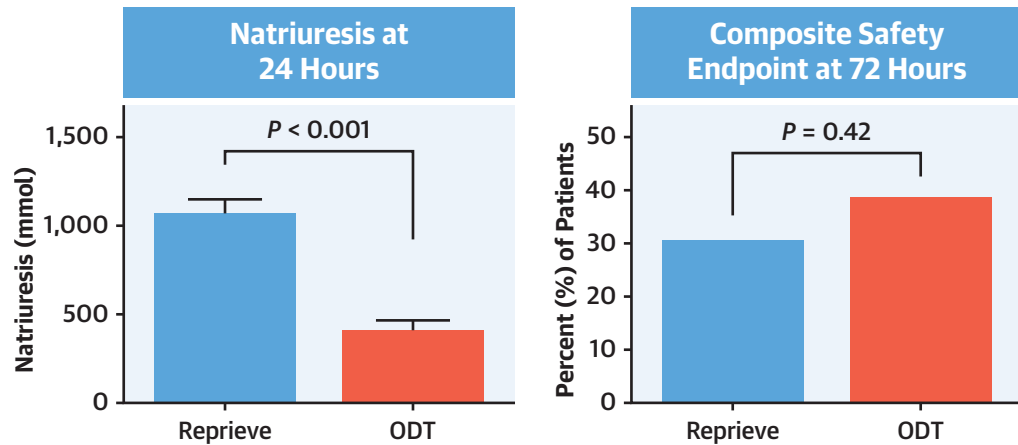
B Time to Adjuvant Thiazide



C Cumulative IV Loop Diuretic during Therapy



(A) Time to loop diuretic titration among patients eligible for further loop diuretic titration. Patients receiving the maximum intravenous (IV) furosemide infusion rate (40 mg/h) after the dose-finding phase are excluded. (B) Time to thiazide addition. (C) Cumulative IV furosemide dose (in milligrams) over the study for Reprieve (blue) and optimal diuretic therapy (ODT) (red) treatment arms.

CENTRAL ILLUSTRATION Primary Efficacy and Safety Endpoints

Udelson JE, et al. JACC Heart Fail. 2026;■(■):103062.

The primary efficacy endpoint of natriuresis 24 hours after randomization is displayed as mean $\pm$ SEM. Primary safety composite endpoint (Kidney Disease: Improving Global Outcomes stage 2 or greater acute kidney injury or use of renal replacement therapy, severe electrolyte abnormality, symptomatic hypotension, or hypertensive emergency) is displayed as the percentage of patients experiencing the safety event. ODT = optimal diuretic therapy

was significantly better compared with ODT (median diuretic efficiency in Reprieve 194 mmol [Q1-Q3: 139-264 mmol] vs ODT 119 mmol [Q1-Q3: 77-151 mmol] of sodium per doubling loop diuretic dose; $P < 0.001$).

The significantly faster rate of decongestion was not accompanied by an increase in safety events. The average increase in serum creatinine from randomization to end of therapy was 0.10 ± 0.27 mg/dL with Reprieve and 0.21 ± 0.33 mg/dL with ODT ($P = 0.09$),

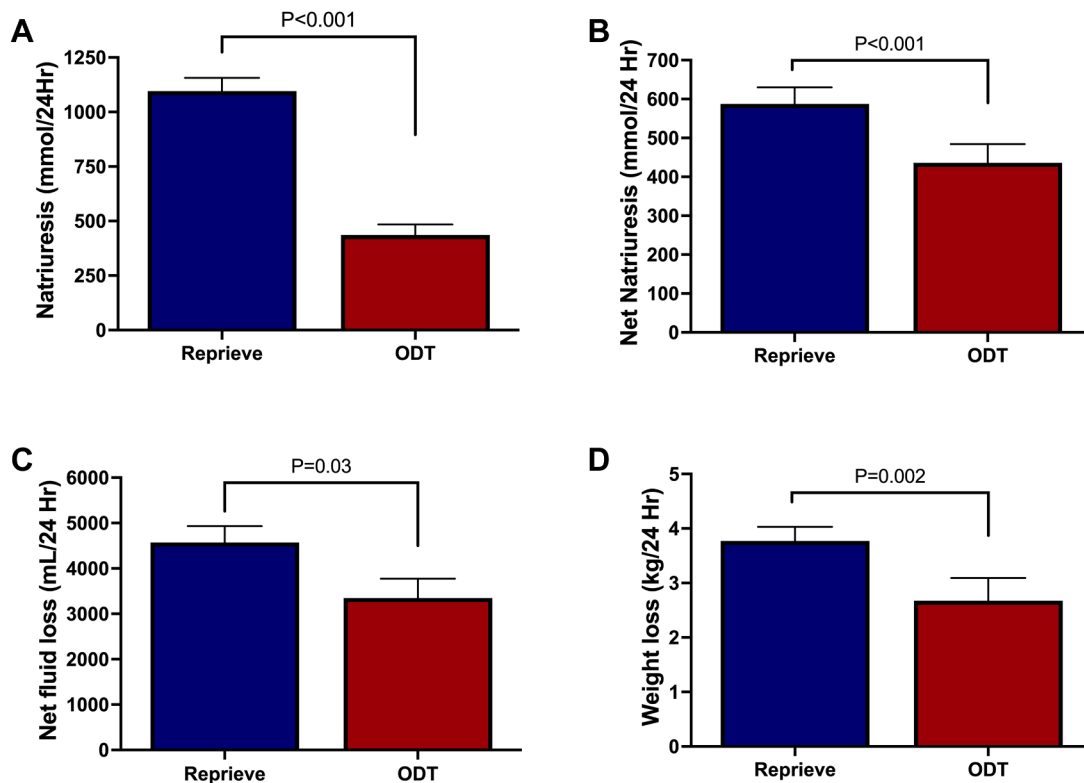
and randomization to peak creatinine was 0.19 ± 0.24 mg/dL with Reprieve vs 0.31 ± 0.39 mg/dL with ODT ($P = 0.066$) (Figure 4). Numerically fewer patients treated with the Reprieve System experienced a serum creatinine increase ≥ 0.3 mg/dL than ODT-treated patients from randomization to peak creatinine (25% vs 34%; $P = 0.33$) and randomization to end of treatment (19% vs 30%; $P = 0.24$). Similarly, the change in serum creatinine from randomization to day 7 or hospital discharge (whichever came first)

TABLE 3 Primary and Secondary Endpoints

	ODT (n = 44)	Reprieve System (n = 52)	P Value
Primary efficacy endpoint (mmol of Na ⁺)	423 $\pm$ 290	1,082 $\pm$ 487	<0.001
Primary safety endpoint (total number of events)	19	16	0.43
Primary safety endpoint (percentage of patients with at least 1 event)	17 (39)	16 (31)	0.42
Serum creatinine >200% or RRT	2 (4.5)	0 (0)	—
Severe electrolyte abnormality	16 (36.4)	16 (30.8)	—
Symptomatic hypotension	1 (2.3)	0 (0)	—
Hypertensive emergency	0 (0)	0 (0)	—
Severe electrolyte abnormalities			
Serum sodium <135 mEq/L	15 (34.1)	13 (25.0)	—
Serum sodium <130 mEq/L	3 (6.8)	1 (1.9)	—
Serum potassium <3.0 mEq/L	1 (2.3)	3 (5.8)	—
Serum magnesium <1.3 mEq/L	0 (0)	0 (0)	—

Values are mean $\pm$ SD or n (%), unless otherwise indicated. Primary efficacy endpoint is the total natriuresis at 24 hours after enrollment.

Na⁺ = sodium; RRT = renal replacement therapy; other abbreviation as in Table 1.

FIGURE 3 Rate of Decongestion

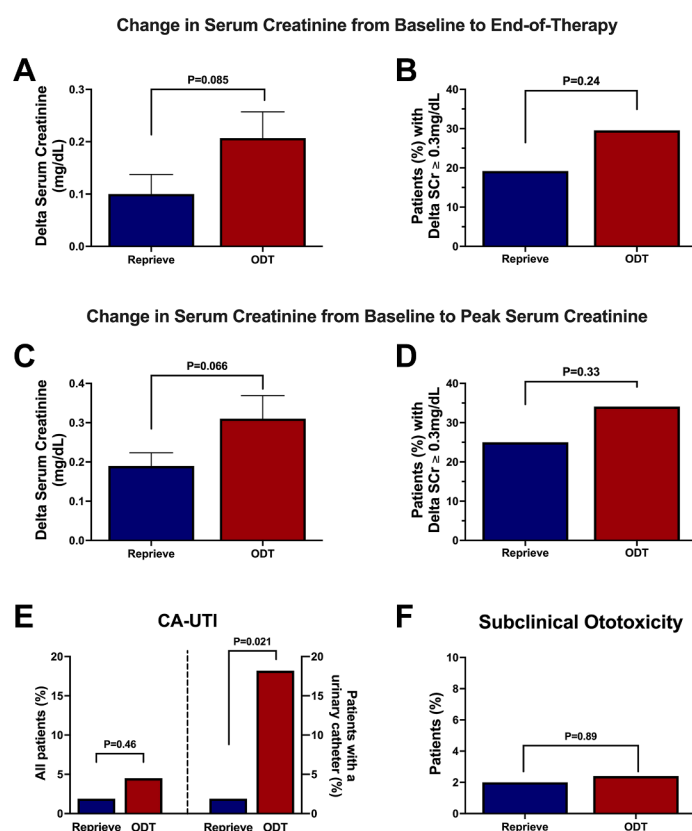
Rate of decongestion per 24 hours of therapy as cumulative natriuresis (A), net natriuresis (B), net fluid loss (C), and weight loss (D) for Reprive and optimal diuretic therapy (ODT). Values are mean $\pm$ SEM.

was 0.11 ± 0.36 mg/dL with Reprive and 0.16 ± 0.35 mg/dL with ODT ($P = 0.54$). Likely due to this absence of kidney- or hypotension-related adverse events, GDMT utilization was maintained despite the rapid decongestion in the Reprive arm (Supplemental Table 3).

The greater IV diuretic exposure in the Reprive treatment arm did not result in ototoxicity. Neither arm had clinically diagnosed ototoxicity adverse events. Subclinical ototoxicity diagnosed by a blinded audiologist using audiometry assessments was uncommon and did not differ between arms from randomization to discharge (Reprive 2.0% [$n = 1$] vs ODT 2.4% [$n = 1$]; $P = 0.890$) (Figure 4). Both treatment arms had similar rates of subclinical shifts in hearing thresholds (Reprive 18% worsened, 22% improved) vs (ODT 17% worsened, 24% improved) according to the ASHA (American Speech-Language-Hearing Association) criteria ($P > 0.78$ between arms).

Despite a higher rate of urinary catheter use in the Reprive arm vs the ODT arm (100% vs 25%), the rate of adjudicated CA-UTI was not higher with Reprive (1.9%) compared with ODT (4.5%; $P = 0.46$) (Figure 4). Comparing only the participants with a urinary catheter in the ODT group ($n = 11$), the rate of CA-UTIs was significantly lower with Reprive therapy ($P = 0.021$). Serious adverse events and treatment- and device-related adverse events did not significantly differ between arms (Supplemental Tables 4 and 5).

Median length of stay was similar between Reprive (5.0 days [Q1-Q3: 2.9-8.2 days]) and ODT (5.2 days [Q1-Q3: 2.8-7.9 days]; $P = 0.97$). Over the 90-day follow-up period, 6 deaths occurred, and 37 patients were rehospitalized for heart failure (Supplemental Table 6). The HR for 90-day death or heart failure hospitalization did not differ significantly between the 2 arms (HR: 0.67 [95% CI: 0.32-1.39]; $P = 0.28$).

FIGURE 4 Safety of the Reprieve System

Change in serum creatinine (SCr) from baseline to the end-of-therapy (A, B) and baseline to the peak value during the 72-hour treatment period (C, D) is displayed as mean $\pm$ SEM and percentage of patients with an increase $\geq$ 0.3 mg/dL for Reprieve and optimal diuretic therapy (ODT). Catheter-associated urinary tract infection (CA-UTI) (E) and subclinical ototoxicity (F) as determined by a blinded audiologist using audiometry are displayed as percentage of patients.

EXPLORATORY ANALYSIS OF THE NATRIURESIS-BASED ALGORITHM. The Reprieve algorithm was changed from a urine volume-based algorithm to a natriuresis-based algorithm approximately halfway through trial enrollment. Although the subgroup size precludes definitive conclusions regarding heterogeneity in treatment effect ($P_{\text{interaction}} \geq 0.051$), exploratory analyses are presented here.

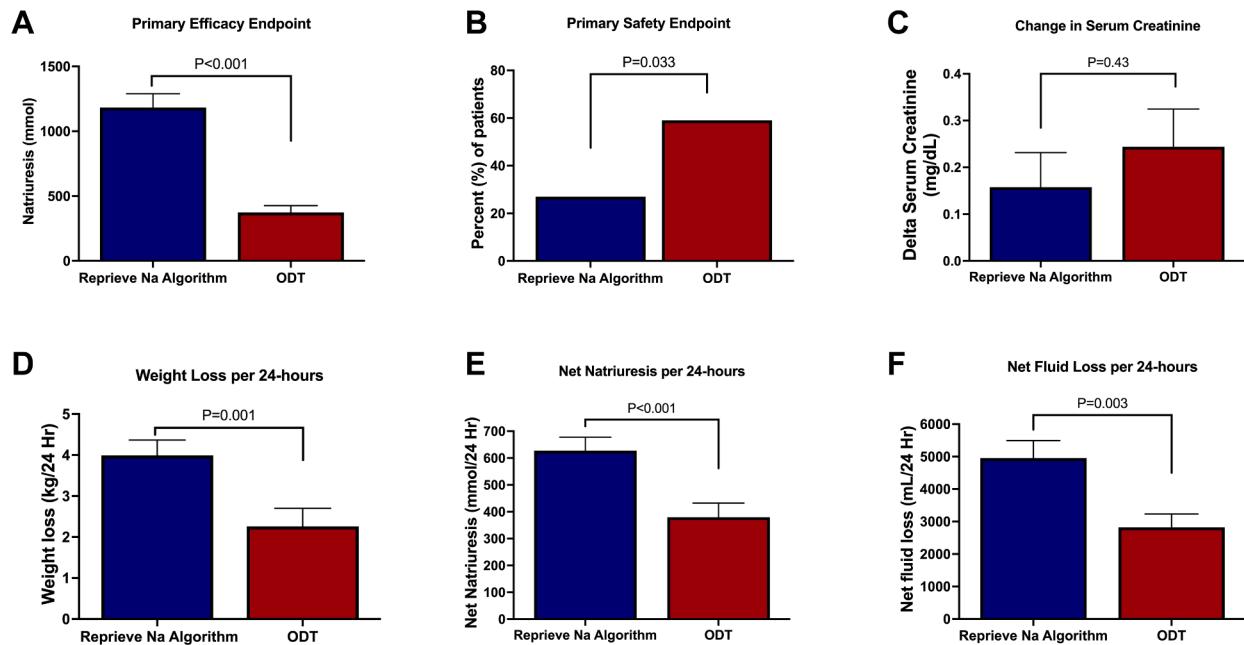
The natriuresis-based algorithm seemed to result in a substantial improvement in the rate of decongestion, but this was not associated with a worsening in safety outcomes or kidney function (Figure 5). These observations can likely be reconciled by the effect on individual patients vs the population level. The mean dose of diuretic, quantity of NaCl replaced, and time on IV diuretic therapy did not differ between the volume and natriuresis-based algorithms

($P > 0.32$ for all). However, the natriuresis-based algorithm produced significantly less variability in these parameters, providing more consistent diuretic ($P = 0.028$) and NaCl ($P < 0.001$) dosing between individual patients (Figure 6). Total decongestion at end of therapy tended to be greater in the natriuresis-based algorithm vs ODT, with greater total weight loss (mean treatment difference 2.5 kg [95% CI: -0.2 to 5.3 kg]; $P = 0.07$), net natriuresis (mean treatment difference 232 mmol [95% CI: -243 to 707 mmol]; $P = 0.33$), and net fluid loss (mean treatment difference 2,426 mL [95% CI: -1,463 to 6,315 mL]; $P = 0.22$). Furthermore, median length of stay tended to improve over ODT with the natriuresis-based algorithm (mean treatment difference 1.7 days shorter with Reprieve [95% CI: -2.0 to 5.5 days]; $P = 0.15$). Details regarding efficacy and safety of the urine volume-based algorithm are provided in Supplemental Figure 3.

DISCUSSION

In this randomized controlled, pilot trial of patients hospitalized with AHF, treatment with the Reprieve System improved the primary efficacy endpoint of 24-hour natriuresis without worsening the primary safety composite endpoint. Secondary and exploratory endpoints consistently indicated significantly more rapid, but safe, decongestion with the Reprieve System. Transition from a urine volume-based algorithm to a natriuresis-based algorithm improved the consistency of therapy for individual patients. These preliminary results informed the design of the ongoing pivotal, multicenter, randomized FASTR II trial (NCT06898515), which will definitively test the hypothesis that the Reprieve System can provide faster and safer decongestion.

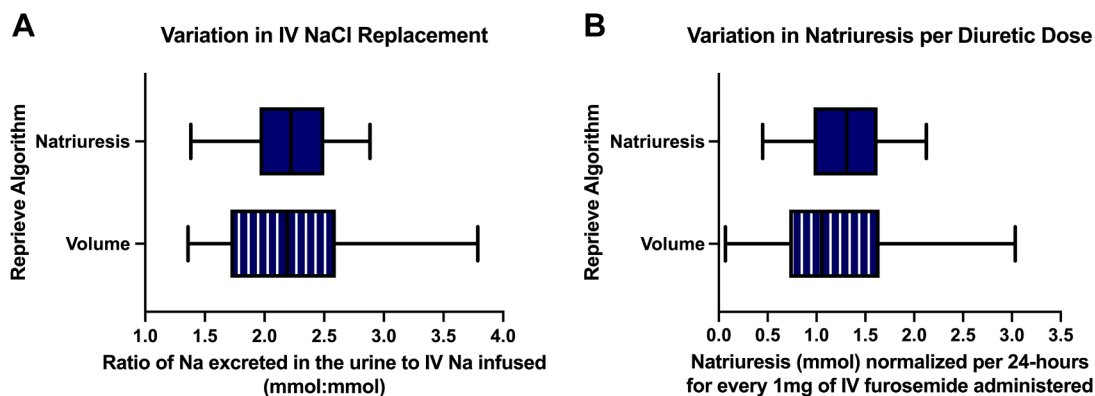
Although direct evidence from randomized trials is currently lacking, incomplete decongestion is consistently associated with increased postdischarge death and rehospitalization.² Regardless of the effect on postdischarge outcomes, rapidly and safely treating congestion remains an obligatory task in the care of patients with AHF. Hospitalized patients with AHF are highly symptomatic, often medically unstable, and most will remain so until decongestion occurs. Faster decongestion could enable earlier discharge or repurpose hospital time toward other diagnostic (eg, amyloid cardiomyopathy evaluation, valve replacement work-up, ischemic evaluation) and therapeutic (eg, GDMT titration, cardioversion, revascularization) activities, which are generally deferred until volume status is improved. In addition, GDMT is critical to improve postdischarge

FIGURE 5 Efficacy and Safety With the Natriuresis-Based Reprive Algorithm

The natriuresis-based (Na) Reprive algorithm is compared with optimal diuretic therapy (ODT) for the primary efficacy endpoint of natriuresis at 24 hours after initiation of randomized therapy (A), the primary safety composite endpoint (B), change in serum creatinine from randomization to end of therapy (C), and the rate of decongestion per 24 hours of therapy defined as weight loss (D), net natriuresis (E), and net fluid loss (F). Values are mean $\pm$ SEM or percentage of patients in B.

outcomes.²⁵ However, worsening kidney function, hypotension, and electrolyte abnormalities that often accompany aggressive decongestion substantially worsen GDMT optimization (~60% increase in

the odds for discharge GDMT discontinuation).²⁶⁻²⁸ Notably, there was no excess of these events in the Reprive arm, and thus despite the rapid decongestion, GDMT utilization was overall similar.

FIGURE 6 Variability in Natriuresis and NaCl Replacement Between Urine Output-Based and Natriuresis-Based Reprive Algorithms

The natriuresis-based (Natriuresis) Reprive algorithm exhibited less variation (distribution width) than the urine volume-based (Volume) Reprive algorithm compared for the ratio of sodium excreted in the urine to the sodium infused as intravenous NaCl (A) and the cumulative natriuresis (millimoles) per 1 mg of IV furosemide over the study period normalized per 24 hours (B). Values are median (Q1-Q3). The F test was used to evaluate the homogeneity of variances ($P < 0.001$ for NaCl replacement and $P = 0.028$ for diuretic dosing).

Therefore, minimizing both hypotension and worsening kidney function remain a clear unmet need for patients with AHF and could enable GDMT optimization during hospitalization to improve postdischarge outcomes.

These observations from the FASTR pilot trial provide encouraging early signals that the Reprive System may be a step toward the goal of rapid and safe decongestion. The overall superiority in the rate of decongestion with Reprive therapy over ODT was substantial, particularly in participants treated with the natriuresis-based algorithm. Importantly, the doses of diuretic agents and the rate of urine production in the ODT arm were similar to those of active therapy arms in the majority of contemporary aggressive decongestion trials ([Supplemental Figure 1](#)). Thus, the difference with Reprive was not due to a less aggressive diuretic strategy in the ODT arm.

Despite this rapid diuresis in the Reprive arm, there was no excess in safety events or deterioration in kidney function. This is in contrast to recent diuretic trials (DOSE, ADVOR, PUSH-AHF [Pragmatic Urinary Sodium-based treatment Algorithm in Acute Heart Failure], and CLOROTIC) in which improvement in decongestion has come at the expense of a substantially increased rate of a ≥ 0.3 mg/dL rise in serum creatinine levels ([Supplemental Figure 1](#)).¹⁵⁻¹⁸ Collectively, observations from these diuretic trials formed the conventional theory that worsening kidney function accompanies more rapid diuresis, essentially in a transactional manner. Beyond the known potent negative effect that worsening kidney function has on GDMT utilization, the direct importance of these small increases in creatinine is unclear.²⁶ Although prior studies found that small increases in creatinine do not offset the benefit of decongestion, it has not yet been determined if achieving the same degree of decongestion without a worsening of kidney function provides benefit because, to date, no therapy has been able to achieve this goal.^{29,30}

Three potential mechanisms may explain the dissociation between the increased rate of diuresis and lack of kidney dysfunction observed with Reprive. First is the rapidity with which decongestive therapy can be initiated and discontinued with the Reprive System. During the initial phase of decongestion, patients have the highest severity of decompensation and are thus most susceptible to in-hospital worsening heart failure. Thus, rapidly achieving successful decongestion, or identifying treatment-refractory patients in need of escalation of care, should facilitate avoidance of in-hospital

worsening heart failure. Importantly, within the first hour of Reprive therapy, a therapeutic diuretic dose is either delivered or high-fidelity diagnostic information is provided identifying a patient resistant to 200 mg furosemide. The end of decongestive therapy, as euvolemia is approached, is also a high-risk time for overdiuresis and worsening kidney function. Most clinicians modify diuretic therapy once a day, barring an adverse event such as hypotension or significant worsening kidney function. Thus, if patients achieve euvolemia, they are generally exposed to continued diuretic therapy until the next day when clinical re-evaluation occurs. Conversely, the Reprive System automatically and rapidly discontinues diuretic therapy when urine output slows if the patient has reached the physician-specified decongestive goals, limiting the potential for hypovolemia-related adverse events.

A second potential difference is the individualized therapy with Reprive. Importantly, in real-world AHF populations, and most trials of decongestive strategies, there is a spectrum of disease severity and diuretic resistance. However, the intervention (eg, increasing loop diuretic agents or adding adjunctive diuretics) is applied to all patients regardless of the need for more intense therapy. With this “one size fits all approach,” participants who would have responded briskly to standard therapy are susceptible to overdiuresis from the unneeded aggressive intervention, precipitating hypovolemia-related adverse events. On the contrary, the Reprive System doses diuretic agents based on real-time diuretic response monitoring, only administering therapy required to achieve the directly measured target diuresis. Moreover, if response becomes excessive, dosing is automatically down-titrated. This tailoring of diuretic dose to the individual patient’s requirement allows very aggressive diuretic dosing when required but avoids aggressive dosing in responsive patients, thus limiting the potential to “overshoot” with diuretic intensity.

A third potential explanation could be due to the Reprive System administering a mean of ~ 3.3 L/day of IV normal saline, which was on average ~ 30 g of NaCl per day. By traditional wisdom this should substantially worsen decongestion; however, diuresis and natriuresis were significantly improved. Importantly, diuretic efficiency, a metric of diuretic resistance specifically querying the loop diuretic resistance independent of dose,³¹ was significantly better with Reprive. This critical finding indicates that the improved natriuresis was not simply the result of administering additional loop diuretic agents to overcome the NaCl administration. Rather,

renal salt-sensing mechanisms were “seeing” the administered NaCl and modulating sodium avidity to improve diuretic resistance and natriuresis. One of the foundational physiological principles of sodium homeostasis is that that NaCl loading of a healthy human will suppress neurohormones and induce natriuresis. More recently, the molecular pathways for renal salt sensing have been characterized.^{12,13} Importantly, emerging evidence suggests that these renal salt-sensing mechanisms remain at least partially intact in heart failure.^{7-11,32} Although administration of 3.3 L/day of normal saline indiscriminately to hospitalized patients with AHF will likely cause harm in many patients, the ability of the Reprieve System to personalize NaCl administration, while simultaneously ensuring a negative sodium balance, provides a unique opportunity to leverage this physiology.

The Reprieve algorithm was transitioned to a natriuresis (sodium output) based algorithm approximately halfway through the pilot trial. Sodium retention is the primary pathophysiology underlying volume expansion, and growing published reports now supports the importance of monitoring urine sodium in the care of patients with heart failure.^{18,19,33} In the majority of patients with heart failure, osmoregulation is intact, and thus the ability to excrete a free water load also remains intact. As such, with significant water intake, a dilute low sodium urine is produced to prevent a decrease in serum sodium. Similarly, when diuretic resistance causes urine sodium concentration to be low, thirst is triggered to increase water intake and prevent hypernatremia. This physiology drives the common discrepancy observed in AHF in which substantial urine volume has been produced with minimal weight loss due to off-setting free water intake. This physiology also explains the current findings related to the change to the natriuresis-based algorithm of more consistent dosing of diuretic and NaCl between patients and improved overall efficacy without worsening safety. With the natriuresis-based algorithm, participants with high urine volume but low urine sodium concentration received more diuretic and less NaCl, whereas patients with high urine volume and high sodium concentration received less diuretic and more NaCl replacement.

Although the primary efficacy endpoint remained positive at 72 hours/end of therapy, there was a catch-up phenomenon noted with ODT resulting in some convergence of efficacy by 72 hours in secondary and exploratory endpoints. This catch-up is expected. In ADHF, there are 2 primary factors determining the amount of decongestion that can be

achieved: first, the between-patient variability in quantity of volume overload at baseline; and second, differences in disease severity and treatment response (eg, diuretic resistance). The amount of volume available for removal is, in general, a much larger driver of quantitative decongestion measures. For example, if 2 patients with 10 lbs of volume overload were randomized to treatment, and 80% decongestion was achieved in control and 100% decongestion with experimental therapy, this would produce 2 lbs' difference between patients. However, if 2 patients were randomized to treatment, one with 10 lbs of volume overload that is 100% decongested (10 lbs loss) and the other with 30 lbs of volume overload that is 50% decongestion (15 lbs loss), the baseline volume overload would dominate the effect. For this reason, the FASTR pilot trial was designed to primarily evaluate early divergence and the rate of decongestion, recognizing that groups will converge over time as more volume-overloaded patients receive additional therapy.

STUDY LIMITATIONS. This was an open-label, intermediate size pilot trial without an a priori hypothesis testing plan, and it is subject to all the limitations inherent to this design. All results should be interpreted as hypothesis-generating. In this pilot, we ensured all patients had decompensated heart failure with substantial volume overload as the primary diagnosis. Many patients were screened for each patient enrolled ([Figure 1](#)), potentially limiting the external validity to real-world practice. This pilot trial was not powered to detect differences in length of stay from faster decongestion, and natriuretic peptide concentration was not available serially to assess that marker of decongestion. Miscellaneous IV fluid volume was not systematically quantified in the ODT arm, potentially overestimating ODT net sodium output. The ODT arm's low loop diuretic dosing in the first hour was driven by the lack of a bolus before starting continuous infusion of the loop diuretic. Although this was the DOSE trial's strategy and a protocolized option for ODT, this approach is arguably suboptimal based upon pharmacokinetic considerations.

The Reprieve algorithm was changed from a urine volume-based algorithm to a natriuresis-based algorithm after an interim analysis, which could have affected the overall results relative to the comparator group. The small subgroups prevent definitive comparisons between algorithms. An additional limitation is that a urinary catheter was not required in the ODT arm, potentially limiting the fidelity of urine collection in this arm. Although the Reprieve System

is considered minimally invasive, the urinary catheter is the primary source of device-related risk, and a key goal of this pilot study was to understand the safety of the device compared with usual care. Furthermore, the design of the Reprise urine collection system and the very brisk urine flow in the Reprise System was expected to substantially reduce infection risk, which is supported by the rates of CA-UTI observed in this trial. Thus, mandating traditional urinary catheter use in the control arm may have provided a disproportionate infection risk and thus, given the aforementioned, was not protocolized. Lastly, this was an open-label trial, in which blinding is difficult because the Reprise System administers large volumes of normal saline. A sham without fluid administration would unblind the care team, and any sham “placebo” fluid would have important safety and physiological effects.

CONCLUSIONS

In this randomized pilot trial of patients hospitalized for hypervolemic AHF, the Reprise System resulted in a faster rate of decongestion without an increased risk of safety events or worsening kidney function relative to ODT. These promising early results have informed the design of the ongoing pivotal FASTR II trial (NCT06898515), which is powered to definitively test the hypothesis that the Reprise System can provide faster and safer decongestion of patients with AHF.

FUNDING SUPPORT AND AUTHOR DISCLOSURES

Dr Testani has received grants and/or personal fees from 3ive labs, Bayer, Bristol Myers Squibb, AstraZeneca, Novartis, Cardionomic, MagentaMed, Reprise Cardiovascular, FIRE1, W.L. Gore, Sanofi, Sequana Medical, Otsuka, Abbott, Merck, Windtree Therapeutics, Lexicon Pharmaceuticals, Precardia, Relypsa, Regeneron, BD, Edwards Lifesciences, Corteria, Terumo, and Lilly; and has a patent for treatment of diuretic resistance issued to Yale and Corvidia Therapeutics Inc, a patent methods for measuring renalase issued to Yale, and a patent Treatment of diuretic resistance pending with Reprise Cardiovascular. Dr Rao has a patent for treatment of diuretic resistance (US20200079846A1) issued to Yale and Corvidia Therapeutics Inc with royalties paid to Yale University, Dr Rao, and Dr Testani; a patent Methods for measuring renalase (WO2019133665A2) issued to Yale; and has received personal fees from Translational Catalyst. Dr Cox has received grants from AstraZeneca; and personal fees from Abiomed, Vectorious, Kestra Medical Technologies, Reprise Cardiovascular, WhiteSwell, and Lexicon Pharmaceutical. Dr Ivey-Miranda has received personal fees from AstraZeneca, Boehringer Ingelheim, Novartis, Merck, and Moksha8.

Dr Udelson has received consulting fees from and/or clinical trial committee work with Reprise CV, Cardurion, Alleviant Medical, FIRE1 Foundry, Merck, Bayer, Cytokinetics, Medtronic, MedTrace, Attralus, and Edgewise. Dr Amin was a site Principal Investigator for the FASTR trial; has participated in clinical trial committee work with Reprise Cardiovascular; and has served as a speaker and/or consultant for Pfizer, Salix, Alexion, AstraZeneca, Bayer, Ferring, Seres, Spero, Eli Lilly, Nova Nordisk, Gilead, Renibus, GlaxoSmithKline, Dexcom, Reprise Cardiovascular, HeartRite, and AseptiScope. Dr Mahler has received funding/support from Abbott Laboratories, Roche Diagnostics, Quidel, Siemens, Grifols, Pathfast, Beckman Coulter, Genetesis, Reprise Cardiovascular, National Foundation of Emergency Medicine, BlueJay Diagnostics, Duke Endowment, Brainbox, the Health Resources and Services Administration (1H2ARH399760100), the Emergency Medicine Foundation, and the Agency for Healthcare Research and Quality (R01HS029017 and R21HS029234); is a consultant for Roche, QuidelOrtho, Abbott, Siemens, Genetesis, Pathfast, and Reprise Cardiovascular; and is the Chief Medical Officer for Impathiq Inc. Dr Gottlieb has served as a consultant to Alnylam Pharmaceuticals, AstraZeneca, Eli Lilly and Company, Johnson and Johnson, and Reprise Cardiovascular; has received speaking and/or lecture fees from Alnylam Pharmaceuticals and Pfizer; and has received grant support from CareDx. Dr Javaheri has a pending patent for fusion protein nanodiscs for the treatment of heart failure and eye disease; is a member of the scientific advisory board of Mobius Scientific; and has received research funding from AstraZeneca and Bitterroot Bio; and has received research grants from Reprise and Alleviant. Dr Damman has received speaker and consultancy fees to his employer by Abbott, AstraZeneca, Boehringer Ingelheim, Novartis, Echosens, and FIRE1. Dr Chung has received research funding from Reprise; has served as a consultant for Medtronic and as a speaker for Boehringer Ingelheim; and is an employee of InterShunt. All other authors have reported that they have no relationships relevant to the contents of this paper to disclose.

ADDRESS FOR CORRESPONDENCE: Dr James E. Udelson, Box 70 Cardiology, Tufts Medical Center, 800 Washington Street, Boston, Massachusetts 02111, USA. E-mail: Judelson@tuftsmedicalcenter.org.

PERSPECTIVES

COMPETENCY IN PATIENT CARE AND PROCE-

DURAL SKILLS: Real-time diuretic monitoring guiding individualized diuretic dose titration and NaCl replacement may allow more rapid decongestion without worsening kidney function.

TRANSLATIONAL OUTLOOK: The encouraging efficacy and safety results of the Reprise System in this pilot trial require further testing in a pivotal trial powered to measure differences in clinical outcomes.

REFERENCES

1. Hollenberg SM, Warner Stevenson L, Ahmad T, et al. 2019 ACC expert consensus decision pathway on risk assessment, management, and clinical trajectory of patients hospitalized with heart failure: a report of the American College of Cardiology Solution Set Oversight Committee. *J Am Coll Cardiol*. 2019;74(15):1966–2011.
2. Chioncel O, Mebazaa A, Maggioni AP, et al. Acute heart failure congestion and perfusion status—impact of the clinical classification on in-hospital and long-term outcomes; insights from the ESC-EORP-HFA Heart Failure Long-Term Registry. *Eur J Heart Fail*. 2019;21(11):1338–1352.
3. Zheng J, Ambrosy AP, Bhatt AS, et al. Contemporary decongestion strategies in patients hospitalized for heart failure: a national community-based cohort study. *JACC Heart Fail*. 2024;12(8):1381–1392.
4. Greene SJ, Triana TS, Ionescu-Iltu R, et al. In-hospital therapy for heart failure with reduced ejection fraction in the United States. *JACC Heart Fail*. 2020;8(11):943–953.
5. Cox ZL, Damman K, Testani JM. Decongestion in heart failure: medical and device therapies. *Nat Rev Cardiol*. 2025;22(12):961–977.
6. Ivey-Miranda JB, Rao VS, Shaburishvili T, et al. Reprieve System for the treatment of patients with acute decompensated heart failure. *J Card Fail*. 2025;31(7):1085–1089.
7. Liu C, Peng Z, Gao X, et al. Simultaneous use of hypertonic saline and IV furosemide for fluid overload: a systematic review and meta-analysis. *Crit Care Med*. 2021;49(11):e1163–e1175.
8. Griffin M, Soufer A, Goljo E, et al. Real world use of hypertonic saline in refractory acute decompensated heart failure: a U.S. center's experience. *JACC Heart Fail*. 2020;8(3):199–208.
9. Biegus J, Iwanek G, Testani J, et al. Sodium chloride versus glucose solute as a volume replacement therapy for more effective decongestion in acute heart failure (SOLVRED-AHF): a prospective, randomized, mechanistic study. *Eur J Heart Fail*. 2025;27(11):2442–2451.
10. Hanberg JS, Rao V, Ter Maaten JM, et al. Hypochloremia and diuretic resistance in heart failure: mechanistic insights. *Circ Heart Fail*. 2016;9(8):e003180.
11. Ter Maaten JM, Damman K, Hanberg JS, et al. Hypochloremia, diuretic resistance, and outcome in patients with acute heart failure. *Circ Heart Fail*. 2016;9(8):e003109.
12. Briggs JP, Schnermann JB. Whys and wherefores of juxtaglomerular apparatus function. *Kidney Int*. 1996;49(6):1724–1726.
13. Subramanya AR, Yang CL, McCormick JA, Ellison DH. WNK kinases regulate sodium chloride and potassium transport by the aldosterone-sensitive distal nephron. *Kidney Int*. 2006;70(4):630–634.
14. Hoorn EJ, Nelson JH, McCormick JA, Ellison DH. The WNK kinase network regulating sodium, potassium, and blood pressure. *J Am Soc Nephrol*. 2011;22(4):605–614.
15. Trulls JC, Morales-Rull JL, Casado J, et al. Combining loop with thiazide diuretics for decompensated heart failure: the CLOROTIC trial. *Eur Heart J*. 2023;44(5):411–421.
16. Mullens W, Dauw J, Martens P, et al. Acetazolamide in acute decompensated heart failure with volume overload. *N Engl J Med*. 2022;387(13):1185–1195.
17. Felker GM, Lee KL, Bull DA, et al. Diuretic strategies in patients with acute decompensated heart failure. *N Engl J Med*. 2011;364(9):797–805.
18. Ter Maaten JM, Beldhuis IE, van der Meer P, et al. Natriuresis-guided diuretic therapy in acute heart failure: a pragmatic randomized trial. *Nat Med*. 2023;29(10):2625–2632.
19. McDonagh TA, Metra M, Adamo M, et al. 2021 ESC guidelines for the diagnosis and treatment of acute and chronic heart failure. *Eur Heart J*. 2021;42(36):3599–3726.
20. 2023 NHSN urinary tract infection (UTI) checklist. Centers for Disease Control National Healthcare Safety Network. Accessed April 16, 2026. <https://www.cdc.gov/nhsn/pdfs/checklists/uti-checklist-508.pdf>
21. Brater DC, Day B, Burdette A, Anderson S. Bumetanide and furosemide in heart failure. *Kidney Int*. 1984;26(2):183–189.
22. Rao VS, Cox ZL, Ivey-Miranda JB, et al. Mechanistic differences between torsemide and furosemide. *J Am Soc Nephrol*. 2024;36(1):99–107.
23. Inker LA, Eneanya ND, Coresh J, et al. New creatinine- and cystatin C-based equations to estimate GFR without race. *N Engl J Med*. 2021;385(19):1737–1749.
24. Rao VS, Planavsky N, Hanberg JS, et al. Compensatory distal reabsorption drives diuretic resistance in human heart failure. *J Am Soc Nephrol*. 2017;28(11):3414–3424.
25. Mebazaa A, Davison B, Chioncel O, et al. Safety, tolerability and efficacy of up-titration of guideline-directed medical therapies for acute heart failure (STRONG-HF): a multinational, open-label, randomised, trial. *Lancet*. 2022;400(10367):1938–1952.
26. Zheng J, Sandhu AT, Bhatt AS, et al. Inpatient use of guideline-directed medical therapy during heart failure hospitalizations among community-based health systems. *JACC Heart Fail*. 2025;13(1):43–54.
27. Ter Maaten JM, Mebazaa A, Davison B, et al. Early changes in renal function during rapid up-titration of guideline-directed medical therapy following an admission for acute heart failure. *Eur J Heart Fail*. 2023;25(12):2230–2242.
28. Gilstrap LG, Fonarow GC, Desai AS, et al. Initiation, continuation, or withdrawal of angiotensin-converting enzyme inhibitors/angiotensin receptor blockers and outcomes in patients hospitalized with heart failure with reduced ejection fraction. *J Am Heart Assoc*. 2017;6(2):e004675.
29. Ahmad T, Jackson K, Rao VS, et al. Worsening renal function in patients with acute heart failure undergoing aggressive diuresis is not associated with tubular injury. *Circulation*. 2018;137(19):2016–2028.
30. Mullens W, Damman K, Testani JM, et al. Evaluation of kidney function throughout the heart failure trajectory—a position statement from the Heart Failure Association of the European Society of Cardiology. *Eur J Heart Fail*. 2020;22(4):584–603.
31. Testani JM, Brisco MA, Turner JM, et al. Loop diuretic efficiency: a metric of diuretic responsiveness with prognostic importance in acute decompensated heart failure. *Circ Heart Fail*. 2014;7(2):261–270.
32. Dauw J, Meekers E, Martens P, et al. Sodium loading in ambulatory patients with heart failure with reduced ejection fraction: mechanistic insights into sodium handling. *Eur J Heart Fail*. 2024;26(3):616–624.
33. Hodson DZ, Griffin M, Mahoney D, et al. Natriuretic response is highly variable and associated with 6-month survival: insights from the ROSE-AHF trial. *JACC Heart Fail*. 2019;7(5):383–391.

KEY WORDS acute heart failure, diuresis, diuretic, heart failure, natriuresis, urine sodium

APPENDIX For an expanded Methods section as well as supplemental figures, tables, and a reference, please see the online version of this paper.